

SAVANA ONLINE E-COMMERCE DATABASE MANAGEMENT SYSTEM

CHAPTER – 1

1.1 Introduction

Savana is an online fashion and lifestyle e-commerce platform that offers a wide variety of products, including clothing, footwear, accessories, beauty products, and other lifestyle essentials. The platform provides customers with a convenient shopping experience through features such as product browsing, secure online payments, order tracking, and doorstep delivery.

The Savana Online E-Commerce Database Management System is designed to manage important information such as customer details, products, sellers, orders, payments, and inventory in a centralized database. The system helps store, retrieve, update, and manage data efficiently while reducing redundancy and maintaining data accuracy. It also supports smooth business operations and improves customer satisfaction by providing quick and reliable access to information.

1.2 Problem Statement

Managing customer records, products, orders, payments, and inventory manually can be time-consuming and may result in errors, duplicate records, and inefficient data management. As the number of customers and products increases, maintaining accurate information becomes difficult without a proper database system.

This project aims to solve these challenges by designing a structured database that stores all business information efficiently. It ensures data consistency, improves security, simplifies order management, and enables faster retrieval of information.

1.3 Objective

The main objective of this project is to design and develop a database for the Savana Online E-Commerce platform that efficiently manages customers, products, sellers, orders, payments, and inventory. The database aims to improve data accuracy, reduce redundancy, maintain data integrity, and support smooth business operations through efficient data management.

1.4 Key Objectives

- To design a relational database for the Savana platform.
- To manage customer and seller information.
- To maintain product and category details.
- To process customer orders and payments.
- To manage inventory and product availability.
- To establish relationships using Primary Keys and Foreign Keys.
- To reduce data redundancy through normalization.
- To ensure secure and efficient data management.

1.5 Business Requirements

The database should support customer registration, product

management, seller management, order processing, payment handling, inventory management, and delivery tracking. It should also maintain customer order history and generate reports for business analysis.

The system should provide accurate data storage, quick retrieval of information, and secure management of customer and business records.

1.6 Stakeholders

The major stakeholders of the project include Customers, Sellers, Administrators, Delivery Partners, and Business Management.

Customers purchase products and place orders, sellers manage products and inventory, administrators maintain the database, delivery partners update delivery status, and management uses reports for business decisions.

1.7 Functional Requirements

The system should provide customer registration and login, product management, category management, seller management, shopping cart, order processing, payment management, inventory management, delivery tracking, customer reviews, report generation, and data retrieval.

These functions help the platform operate efficiently and provide a better shopping experience.

1.8 Non-Functional Requirements

The system should provide high performance, security, reliability, scalability, and data integrity. It should retrieve data quickly, protect customer information, and support future expansion.

The database should also be easy to maintain and ensure regular backup and recovery of data.

1.9 Motivation to the Project

The motivation behind this project is to understand how a real-world fashion e-commerce platform like Savana manages large amounts of customer, product, seller, and order data using a Database Management System. It provides practical knowledge of database design, SQL, normalization, and table relationships.

This project also helps improve database management skills while demonstrating how an efficient database supports smooth business operations, reduces manual work, and enhances the overall online shopping experience for customers. This version matches the same style and length as your Shopsy report, making it suitable for a college DBMS mini-project report.